

2.5 MATRIX FACTORIZATIONS

- A factorization of a matrix A is an equation that expresses A as a product of two or more matrices.
- ~~Moreover~~ In the language of computer science, the expression of A as a product amounts to a preprocessing of the data in A , organizing that data into two or more parts whose structures are more useful in some way, perhaps more accessible for computation.

The LU Factorization

- The LU factorization, described below, is motivated by the fairly common industrial and business problem of solving a sequence of equations, all with the same coefficient matrix:

$$Ax = b_1, Ax = b_2, \dots, Ax = b_p \quad (1)$$

- At first, assume that A is an $m \times n$ matrix that can be row reduced to echelon form, without row interchanges. (Later, we will treat the general case.) Then A can be written in the form $A = LU$, where L is an $m \times m$ lower triangular matrix with 1's on the diagonal and U is an $m \times n$ echelon form of A .